RADIATION

Detailed Student Notes - Part 1 & Part 2

PART 1 - INTRODUCTION TO RADIATION

Radiation is the transfer of energy through electromagnetic waves. Unlike conduction and convection, radiation does not require a material medium.

Radiation -> transfer of energy by electromagnetic waves

No material medium is required.

1. Fundamental facts about radiation

1. Every body above absolute zero (0 K) emits radiation continuously.
2. Radiation does not require a medium. It can pass through vacuum and electromagnetic radiation travels at the speed of light.
3. A body can continue to emit radiation even when it is at the same temperature as its surroundings.

2. Emission and absorption occur simultaneously

A body does not only emit radiation. It also receives and absorbs radiation coming from the objects and surroundings around it.

If the body and its surroundings are at the same temperature, the body still emits radiation, but it absorbs radiation at the same rate.

Energy emitted per second = Energy absorbed per second

Therefore, net heat transfer by radiation = 0.

Important: Zero net heat transfer does NOT mean that radiation has stopped. Emission and absorption are still taking place continuously; their rates are equal.

3. What "net radiation" means

For thermal exchange, what matters is not simply whether a body is emitting radiation, but whether the amount emitted is greater than, equal to, or less than the amount absorbed from the surroundings.

Situation	Result
Emission = absorption	No net heat transfer
Emission > absorption	Net energy leaves the body
Emission < absorption	Net energy enters the body

PART 2 - ABSORPTION, EMISSION, EMISSIVITY & BLACK BODY

1. What happens when radiation falls on a surface?

When electromagnetic radiation is incident on a surface, the incident energy can be divided into three parts:

- a part is absorbed by the surface,
- a part is reflected by the surface,
- a part may be transmitted through the surface.

If the incident energy is taken as the whole amount, the fractions absorbed, reflected and transmitted must add up to 1.

$$a + r + t = 1$$

Symbol	Quantity	Meaning
a	Absorptive power	Fraction of incident radiation absorbed
r	Reflectance	Fraction of incident radiation reflected
t	Transmittance	Fraction of incident radiation transmitted

2. Absorptive power, a

Absorptive power is the fraction of the incident radiation that is absorbed by the surface.

$$a = \text{Energy absorbed} / \text{Energy incident}$$

Absorptive power is dimensionless and 0 <= a <= 1.

Meaning of the limiting values:

- a = 0: none of the incident radiation is absorbed.
- a = 1: all of the incident radiation is absorbed.

Example: if a = 0.8, the surface absorbs 80% of whatever radiation falls on it.

3. Emissive power, E

Every body above 0 K emits radiation. Emissive power tells us the amount of energy radiated by a surface per unit area per unit time.

$$E = \text{Energy radiated} / (\text{Area} \times \text{Time})$$

SI unit: W/m^2

- Emissive power is an actual rate of radiation per unit area.
- It has units.
- It depends strongly on the temperature of the body.

4. Emissivity, e

Emissivity describes how good a surface is at emitting radiation compared with a perfect emitter at the same temperature.

e = E / E_black body (at the same temperature)

0 <= e <= 1; emissivity has no unit.

If e = 0.6, the surface emits 60% of the radiation that a perfect emitter would emit at the same temperature.

Do not confuse: Emissive power E is an actual quantity with unit W/m^2. Emissivity e is only a dimensionless comparison number.

5. Emissive power and emissivity - comparison

Feature	Emissive power, E	Emissivity, e
Meaning	Energy radiated per unit area per unit time	Quality of a surface as an emitter
Unit	W/m^2	No unit
Range	Not restricted to 0-1	0 to 1
Temperature	Depends strongly on temperature	Used as a surface property for this course

6. Perfectly black body

A perfectly black body is an ideal body that absorbs all radiation falling on it, irrespective of the incident radiation.

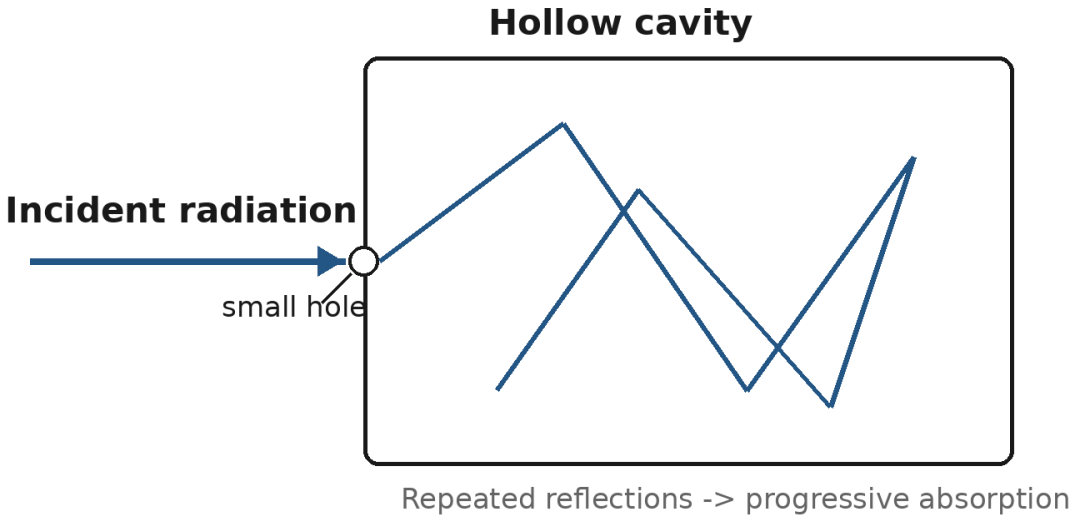
For a perfectly black body: a = 1, r = 0, t = 0

It is also the best possible emitter, so its emissivity e = 1.

The fact that the best absorber is also the best emitter is explained by Kirchhoff's law, which comes in the next part of the lecture.

7. Practical black body - cavity with a small hole

A hollow cavity with a very small opening behaves approximately like a perfect black body.



4. Radiation enters through the small hole.
5. It undergoes repeated reflections from the inner walls of the cavity.
6. At every reflection, some of the radiation is absorbed.
7. After many reflections, almost all of the incident radiation is absorbed and very little finds its way back out through the small hole.

Key point: The small HOLE behaves like the black body. The entire box is not the black body.

8. Why does the small opening look black?

Very little of the radiation entering the opening returns directly to the observer. Most of it undergoes repeated reflections and is absorbed inside the cavity. Hence the opening appears dark.

The lecture uses the pupil of the eye and a window of a house viewed from far away in daylight as familiar examples of openings that can appear dark for the same reason.

9. "Black body" does not mean "black-coloured body"

The word black in black body refers to perfect absorption of radiation. It does not mean that the body must always appear black to our eyes.

A black body at a sufficiently high temperature can emit visible radiation and may glow brightly. A very hot black body can appear red, yellow, or white.

JEE trap: "The emissive power of a perfectly black body is 1" is false. It is the emissivity e that equals 1. Emissive power E has unit W/m^2 and depends on temperature.

10. Part 1 & Part 2 - final revision

- Radiation is transfer of energy through electromagnetic waves and does not require a material medium.
- Every body above 0 K emits radiation continuously.
- At thermal equilibrium, emission and absorption continue but are equal, so net heat transfer is zero.
- When radiation falls on a surface: a + r + t = 1.
- Absorptive power a is a dimensionless fraction between 0 and 1.
- Emissive power E is energy radiated per unit area per unit time; its unit is W/m^2.
- Emissivity e compares a surface with a perfect emitter at the same temperature; 0 <= e <= 1.
- A perfectly black body has a = 1 and e = 1.
- A small hole in a cavity is the practical model of a black body.
- Black body refers to absorption properties, not necessarily visible black colour.